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Computer Science and Engineering Department
M.Tech(CSE/DS)-I - Sem-I - 2026-27
Computer Vision and Image Processing (CSCS111/CSDS119)
Lab Assignment 4: Statistical Order Filtering

Note: For this experiment, the Cameraman image will be used.

1. Write a program to show the working of the statistical order filtering.

- The goal of this project is to implement and analyze order statistics filtering techniques for image processing, specifically focusing on noise reduction and feature enhancement. Order statistics filters, such as the median filter and max/min filters, midpoint, alpha-trimming operate by sorting pixel values within a neighborhood and selecting specific values (e.g., median, minimum, or maximum) to replace the central pixel, making them effective for removing noise while preserving edges.
- Implement other filters that replace the central pixel with different ranked values (e.g., 25th percentile or 75th percentile) to explore the effect of selecting different statistical ranks in a neighborhood.

2. Apply each filter to the noisy Cameraman image and display the filtered images. Compare the results for each filtering method, focusing on noise reduction and the preservation of edges and details.

- Experiment with different neighborhood sizes (e.g., 3×3 , 5×5 , 7×7) and assess how the filter size impacts the trade-off between noise reduction and image detail preservation. Write your observation.